

ml-algo

Using machine learning to find stock-trading strategies that really work, tested honestly on real market data

In one sentence: we are building a daily-chart system that picks stocks the way top momentum traders do (Quillamagie, Minervini, O'Neil), and we use machine learning and strict testing on 20 years of real data to find out which of their rules actually make money, and which only look good in hindsight.

The bottom line so far

1. A real but modest edge exists, mostly from three things: owning the strongest stocks (momentum), buying stocks that gap up on earnings news (episodic pivots), and sizing positions by how volatile they are.

2. Many popular trading rules do NOT add anything on real data: chart patterns (VCP, triangles, flags, tightness), candle patterns, market-timing filters (21/50 averages, breadth, A/D line), hot sectors, and predicting next-day / next-week direction (a coin flip).

3. Our biggest lesson was survivorship bias. Testing only on today's stocks made results look about twice as good as reality. After adding stocks that later dropped out of the S&P 500, our ML stock-picking model made 10.5% a year vs 14.6% for simply holding SPY. A simple momentum portfolio made 22% a year (but with deep drawdowns).

4. Latest (runs 25-26): a concentrated 3-5 stock leader portfolio on top traders' rules works best with a setup entry, a 50-day exit and a QQQ market filter, and requiring sales growth 20%+ and EPS growth 25%+ improved it in 83% of versions. But it still did not beat SPY, QQQ or a simple 10-stock momentum portfolio. Next: a diversified 10-20 stock momentum book with the fundamentals rule, then 3-6 months of paper trading.

Progress at a glance

Part	Status	Where it stands
Finding what works (research)	~80% done	26 research runs; the main questions are answered
Honest testing (no hidden bias)	~75% done	Survivorship mostly fixed; fully clean data still missing
Final strategy design	In progress	Fundamentals rule found (run 26); diversified version next (run 27)
Daily tool (today's picks)	Built, needs upgrade	Works as promised in a replay (42% hit rate) but uses an older model
Proof in real time (paper trading)	Just started	60 picks logged; needs 3-6 months

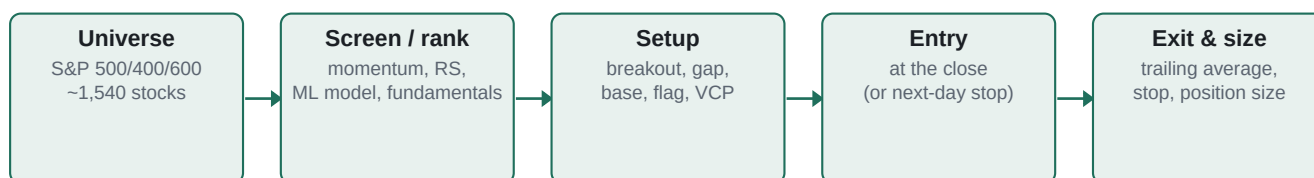
1. What we are trying to do

The goal, in the user's own terms:

- **Daily chart only**, no intraday trading.
- Follow the style of top momentum traders: buy strong leading stocks breaking out or gapping up on news, cut losers fast, let winners run.
- Catch big movers like NVDA, SNDK or PLTR **once they are going up** (no need to be early), with the fundamentals backing the move.
- Original target: a high chance of +10% to +20% before a -10% loss. (We learnt that 'high chance' is set by the bracket, not by skill: see page 6.)
- Hold a small, concentrated portfolio (3-5 stocks) that updates every day from the closing prices.

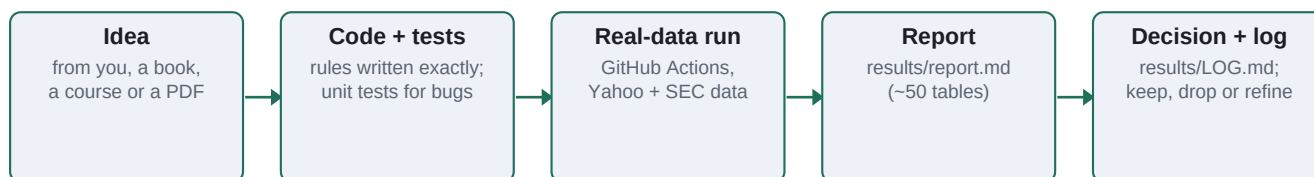
How a strategy is built

Every strategy we test follows the same chain. Each box is something we test many versions of:



How the research works

We never trust a rule because a famous trader uses it. Each idea goes through the same loop:



Each research run takes about 1-1.5 hours on GitHub's servers, downloads 20 years of daily prices, tests hundreds of thousands of trades, and writes its findings back to the repository. 26 runs have been done.

2. How we keep the testing honest

Most trading results online are too good to be true because of a few common mistakes. These are the rules we follow to avoid them:

Rule	Why it matters
Real data only (Yahoo prices, SEC filings)	Made-up data can't tell us anything about real markets.
Choose on 2006-2017, judge on 2018-today	If you pick the best rule using all the data, it will look great by luck. We pick on the first period and check on the second, which the choice never saw.
Compare with random entries	A rule only has an edge if it beats buying random stocks in an uptrend with the same exit. Rising markets make almost everything look good.
Point-in-time stocks, incl. former members	Testing only on today's S&P 500 hides the stocks that crashed and were removed (survivorship bias). This inflated our results about 2x.
No peeking at the future	Every signal uses only data known at that day's close; filings count from the day after they were filed. Automated tests check this.
Realistic costs	0.1% per side by default, stress-tested at 0.3%.
Write the rule down before testing	Rules from a source are tested exactly as written, and what would prove them wrong is decided in advance.
Ask 'who is overpaying, and why?'	A real edge needs a counterparty with a reason (slow funds, forced sellers, underreaction to news). Indicators with no reason behind them usually fail.

Key terms used in this document

Term	Meaning
R	Profit measured in units of the risk taken. +1R = made as much as was risked; -1R = lost the full risk.
IS / OOS	In-sample (2006-2017, used to choose) and out-of-sample (2018-today, used to judge).
Point-in-time (PIT)	Only counting a stock while it was actually in the S&P 500, including stocks later removed. The honest benchmark.
CAGR / max drawdown	Average yearly growth / the worst fall from a peak.
ADR	Average daily range: how much a stock typically moves in a day (5% = volatile).
Episodic pivot (EP)	A stock gapping up 8-10%+ on very high volume, usually on earnings news.
Momentum / RS	How strongly a stock has risen vs others over 1-12 months.

3. The honest scoreboard

Portfolio results from 2018 to today, using only stocks that were in the S&P 500 at the time (including ones later removed), 0.1% costs per side. This is the fairest comparison we can make with free data.

Approach	Return / year	Worst drawdown	Verdict
Simple momentum portfolio (top 10 by 6m/1m return, monthly)	22.4%	-41%	Best return, deep drops
ML stock-picking model + size by volatility (ADR)	16.3%	-32%	Beats SPY
SPY (S&P 500 index fund)	14.6%	-34%	The bar to beat
Episodic pivots (earnings gaps), strong stocks, 50-day exit	13.0%	-25%	Lower risk
3-5 stock leader portfolio, trader rules (run 25 best)	15.3%	-29%	About SPY; weak before 2018
Same + sales/EPS growth required (CAN SLIM style, run 26)	11.2%	-26%	Lower drawdown
ML stock-picking model, equal size	10.5%	-33%	Doesn't beat SPY
Same model, costs 0.3% per side	7.0%	-41%	Costs matter
Breadth timing of SPY (course idea)	5.2%	-28%	Fails
The workflow PDF's full plan, as written	6.3%	-17%	Too small positions

Note: in earlier runs the ML model showed 21-28% a year. Those numbers were inflated by survivorship bias (page 6). Numbers on all stocks without the point-in-time rule are still higher (about 27%) and should not be trusted.

Per-trade edges that held up in both periods

Setup or rule	Before 2018	2018-today	Random entries (same rules)
Episodic pivot on an earnings day (50-day exit)	+0.28R	+0.44R	about 0R
Rocket breakouts (6-week base to new high / 5% gap holding)	+0.06 to +0.07R	+0.13 to +0.20R	-0.14 / -0.06R
21/50 EMA cross then base breakout, strong stocks (course)	+0.10R	+0.21R	-0.13 / +0.01R
Ranking same-day signals by relative strength		about +16 points a year vs random order	

4. What works and what doesn't

What held up on real data

Finding	Evidence	Who's overpaying (the reason)
Momentum: own the strongest stocks	Momentum portfolio 22%/yr PIT; RS filters help every setup	Fund money arrives slowly; investors underreact
Episodic pivots (gap up on earnings)	+0.44R on earnings days vs +0.10R for other gaps	Institutions take weeks to rebuild positions after a surprise
Buy the fear	Model picks hit +20% first 43% of the time when VIX is 20-30 vs about 30% when calm	Forced sellers (margin calls, volatility-targeting funds)
Close below the 50-day average as the exit	Most robust exit in every run	Lets big trends run
Size by volatility (ADR) / adapt size to recent results	PIT 16% vs 10.5%; adaptive sizing helped every time	Risk control
QQQ above its 200-day for breakouts	Breakouts +0.2R in green markets, negative otherwise	Breakouts need buyers

What did NOT hold up

Idea	Result
Chart patterns: VCP, flags, ascending triangles, tight coils, Darvas-style bases	No edge over random entries
Candles, gaps, fair-value gaps, round numbers, support/resistance for next-day direction	Coin flip (51-52%)
Market filters: SPY/QQQ vs 21/50 averages, breadth, A/D line	Didn't raise the odds; mostly cut good trades
Leading sectors / sub-industries (both strong, both green, top 3)	Cut returns roughly in half
Up/down volume 'accumulation', not chasing 10% above the 21 EMA	No help
Fundamental 'inflection' (accelerating sales, margins, FCF) as a filter	No help as a filter; small help as an ML input
Qullamaggie's fast 10/20-day exits on daily bars	Lose money (his intraday entries can't be tested here)
Supertrend + 21 EMA (claimed 60-70% win rate)	29-47% win rate, small edge
Parabolic shorts	60% win rate but about 0% average return
A 70% win rate on +20/-10	Not realistic: see page 6

5. The three biggest lessons

Lesson 1: survivorship bias can double your results

If you test on today's S&P 500 members, you only test companies that survived and grew. The ones that crashed were removed and are missing. In run 12, the same strategy made 47% a year on all dates but only 20% when a stock was traded only after it actually joined the index. In run 24, adding 158 former members cut the ML model from 21% to 10.5% a year.

Same ML model, 2018-today	Return / year	Worst drawdown
Today's members, all dates (run 18)	21.3%*	-30%
Current members only, point-in-time (run 24)	12.9%	-26%
Including former members (run 24, honest)	10.5%	-33%
SPY	14.6%	-34%

* Run 18 used today's member list. Stocks that bankrupted or were bought out are still missing, so even 10.5% may be slightly optimistic.

Lesson 2: win rate is set by the bracket, not by skill

A stock moving randomly hits +20% before -10% about a third of the time, because the target is twice as far as the stop. Our best picks did this 38-44% of the time. You can get a 70%+ win rate with +5/-10, but you lose twice as much each time you're wrong, so it earns the same or less. Top traders like Quillamaggie win only 25-30% of their trades and make their money on a few big winners.

Target / stop	Our hit rate	Break-even	Profit per trade
+5% / -10%	72%	67%	+0.8%
+10% / -10%	57%	50%	+1.5%
+20% / -10%	38%	33%	+2.6%
+20% / -15%	46%	43%	+4.0%

Lesson 3: our ML edge was mostly risk, not skill

Compared with random stocks of the same volatility and momentum, the ML model's picks earned only about +1% more per trade after 2018 and about 0% before; the edge was positive in only 10 of 19 years. After adjusting for market exposure, its extra return over SPY (+1.5% a year) was not statistically meaningful. In plain terms: the model mostly learnt 'volatile momentum stocks go up more in rising markets', which you are paid for by taking more risk. That is why we moved to a simpler, rule-based leader portfolio.

6. Ideas tested from the user's sources

Every source was turned into exact rules and tested, not taken on trust:

Source	What was tested	Verdict
Qullamaggie (breakouts, EPs, parabolic shorts)	His scan (top 3% performers, ADR 4%+), flag breakouts, EPs, 10/20-day trailing exits, QQQ regime, parabolic shorts	Scan + EPs work; fast exits fail on daily bars; shorts no edge
Minervini / O'Neil (VCP, CAN SLIM)	Trend template, VCP, bases, 7-8% stops, 50-day exit, 52-week highs	50-day exit and stops work; VCP itself no edge
Your multibagger / playbook notes	8/21 EMA cross, retests, undercut, flags, trim exits	Mostly no edge; RS filter helps
Stock Selection Workflow PDF	Rockets, Nasdaq-100 scanner, regime, sizing, 40/25/15/20 split	Rockets work; scanner doesn't; plan as sized made 6%/yr
Early fundamental inflection (video)	Accelerating sales, operating leverage, margins, FCF, up/down volume, not extended	No help as filters; small help inside the model
ADR% infographic	5% minimum, 5-12% sweet spot, \$10M liquidity, size by ADR	Sizing by ADR helps; buckets don't
Techno Charts swing course (79 videos)	21 EMA rules, breadth timing, momentum portfolio, Supertrend, EMA-cross bases, reversals	Momentum portfolio and EMA-cross bases work; most else fails
'Who is overpaying?' framework	Earnings placebo, volatility-matched check, survivorship	Confirmed EPs; exposed the ML model
Short-term direction (green day, 3 days, week)	Candles, gaps, FVGs, levels, VIX, weekly structure	Coin flip (AUC 0.51-0.52)

7. Timeline of the research runs

Run	Question	Answer
1-3	Which classic setups beat random entries?	Episodic pivots (all variants); the 50-day exit; ranking by RS
4-6	Selection method, themes, honest drawdowns, patterns	Pick by expected R; themes and most patterns don't help; drawdowns ~50%
10-12	How do top traders beat the market?	Stock selection model (4x hit rate) but survivorship inflates results
13-14	Your goal: +10/20% before -10%	38-44% hit rate vs 33% break-even; first daily picks tool
15	Replicate Qullamaggie	Scan and EPs work; fast exits fail on daily bars
16	Green day / next 3 days / next week	Coin flip
17-18	Bracket menu, market regime, VIX	Win rate set by geometry; buy the fear
19, 21	Leading sectors, 21/50 averages, breadth, A/D line	None help
20	Your workflow PDF	Rockets work, scanner doesn't, sizing too small
22-24	Fundamentals, ADR, earnings, survivorship, course ideas	Big survivorship correction; momentum + EP + ADR sizing hold up
25-26	3-5 stock leader portfolio on traders' rules + fundamentals	Setup + 50-day + QQQ filter works; sales/EPS growth helps; still not above SPY

8. Latest results and what's next

Runs 25-26: a concentrated leader portfolio on top traders' rules

- Hold **3 to 5 leaders** (rising 50-day above the 200-day, near 52-week highs, ranked by momentum), equal weight, decided at each close; 1,008 versions tested, picked on 2007-2017, judged on 2018-today.
- **What works:** buy only on a setup (EP, breakout, base, flag), exit on a close below the 50-day, and pause new buys while QQQ is below its 200-day. Fast 10/20-day exits and 'any leader' rotation fail (-50% to -70% drawdowns).
- **Fundamentals help:** requiring sales growth 20%+ and EPS growth 25%+ (from SEC filings, known at the time) improved 83% of versions after 2018: median 6.6% to 12.3% a year, drawdown -46% to -34%.
- **But concentration costs return:** the best versions made 11-15% a year after 2018 vs SPY 14.6%, QQQ 20.2% and a simple 10-stock momentum portfolio 22.4%.
- **NVDA / PLTR-type stocks:** caught in pieces (PLTR +31%, ANET +35%, NVDA +10%), often after one or two -8% stop-outs. A fixed 8% stop is too tight for such volatile leaders.

Tools already built

Tool	What it does	Status
Research pipeline (research.py)	Runs every test on GitHub and writes results/report.md	Working
Daily picks (picks.py)	Ranks stocks each evening with stops and targets	Works (42% hit rate in replay); old model
Forward test log	Records the daily top 10 and scores them as prices arrive	60 picks logged, none finished yet

Next steps

#	Step	Why
1	Run 27: 10-20 stock momentum book + sales/EPS rule + 50-day exit + wider ADR-based stop	Combine what held up; judge from 2010 when filings exist
2	Turn the daily picks into a 'today's portfolio' list	Hold these, sell when this level breaks
3	Merge to the main branch	So the daily list and paper trading run automatically every evening
4	Paper trade for 3-6 months	The only test the strategy has never seen
5	Verify on survivorship-free data (QuantConnect or paid data)	Close the last big source of bias

Open decisions

- **Merge to main** (needed for the automatic daily run): waiting for the final strategy.
- **Data:** free data so far. QuantConnect (delisted stocks included) needs a new session with that website allowed. Alpha Vantage's free tier (25 requests a day) is too small for full tests; a premium key would add earnings surprises and some delisted prices.
- **Singapore:** not tested yet; Yahoo has SGX prices, but announcement dates are harder to get.

9. Honest limits

- **Daily bars only:** intraday entries (Qullamaggie's opening-range high with a stop at the day's low) can't be tested, and that may be where some of his edge comes from.
- **Survivorship is reduced, not removed:** stocks that went bankrupt or were bought out have no free prices.
- **2018-today has now been looked at many times.** Anything chosen because it looked good after 2018 is not proven by it. Paper trading is the real final test.
- **Concentrated portfolios swing hard.** With 3-5 stocks at 20-33% each, one bad gap can cost 5-10% of the account. Drawdowns of 30-40% are realistic even for good versions.
- **This is research, not financial advice.** Nothing here guarantees future results.

10. Where everything lives

File / folder	What's in it
results/report.md	The latest full research report (all tables)
results/LOG.md	Every run: question, change, result, decision (newest first)
results/today_picks.md, forward_test.md	Daily picks and the paper-trading record
CLAUDE.md	Project rules and the list of known findings
mlalgo/research/	The code: entries (setups), engine (exits), analyze (tests), rotation (portfolio), fundamentals (SEC data), shorts
research.py, picks.py	Run the research / make today's picks
.github/workflows/	Automatic runs on GitHub (research on every code change, picks daily)